3.3.2 Division of an algebraic expression (polynomial) by a monomial

To divide a polynomial by a monomial, divide each term of the polynomial by the monomial.

Example 3.7

Divide : $(5y^3 - 25y^2 + 8y)$ by $5y$

Solution:

We have, $(5y^3 - 25y^2 + 8y) \div 5y$

$$= \frac{5y^3 - 25y^2 + 8y}{5y}$$

$$= \frac{\cancel{5}y^3}{\cancel{5}y} - \frac{\overset{5}{25}y^2}{\cancel{5}y} + \frac{8y}{5y}$$

$$= y^{3-1} - 5y^{2-1} + \frac{8}{5}$$

$$= y^2 - 5y + \frac{8}{5}$$

Think

Are the following divisions correct?

(i) $\frac{4y+3}{4} = y + 3$ (ii) $\frac{5m^2+9}{9} = 5m^2$

(iii) $\frac{2x^2+8}{4} = 2x^2 + 2$. If not , correct it.

Try these

(i) $(16y^5 - 8y^2) \div 4y$

(ii) $(p^5q^2 + 24p^3q - 128q^3) \div 6q$

(iii) $(4m^2n + 9n^2m + 3mn) \div 4mn$

3.4 Avoid Some Common Errors

S.No	Error	Correct	Reason
1.	$2xx = 2x$	$2xx = 2 \times x^1 \times x^1 = 2x^2$	Product of variables
2.	$-3x - 4x = -1x$	$-3x - 4x = -7x$	Same sign factors should be added and put the same sign.
3.	$4y + 3y + y = 7y$	$4y + 3y + y = 8y$	y is same as $1y$,co-efficient 1 of a term is usually not written.
4.	$5x + 3x = 8x^2$	$5x + 3x = 8x$	When we add or subtract like terms, add or subtract only the co-efficient of the like terms, keep the variable as it is.
5.	$9x + 1 = 10x$	$9x + 1 = 9x + 1$	Unlike terms cannot be added or subtracted
6.	$3x + 4y = 7xy$	$3x + 4y = 3x + 4y$	Unlike terms cannot be added
7.	$3(4x + 9)$ $= 12x + 9$	$3(4x + 9) = 12x + 27$	3 is common factor multiply both the terms.
8.	$5 + (3y - 4)$ $= 15y - 20$	$5 + (3y - 4)$ $= 5 + 3y - 4$	Addition symbol is in between the terms, not multiplication

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